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REGULAR PAPER

Infrared small-target detection based on multi-directional multi-scale high-boost response

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Abstract

As of late, infrared (IR) small-target detection technology is broadly utilized in low-altitude monitoring frameworks, target-tracking frameworks, precise guidance frameworks and forest fire prevention frameworks. In this paper, we propose an infrared small-target detection strategy based on multi-directional multi-scale high-boost response (MDMSHB). First, an eight-direction filtering template is proposed, which can consider the directional information of the image and significantly suppress heterogeneous background such as cloud, linear interference and interface like ocean–sky background. Then, a map based on multi-directional multi-scale high-boost response (MDMSHB map) is calculated. Finally, a straightforward threshold segmentation technique is utilized to get the detection result. The simulation results comparing this method with the four state-of-the-art strategies in six sequences demonstrate that the proposed strategy can adequately suppress heterogeneous background and arbitrary noise. The approach can improve detection rate and reduce false alert rate as well.

Keywords Detection · Infrared small targets · Directional filters · High-boost response · Human visual system

1 Introduction

As of late, infrared (IR) small-target detection technology is broadly utilized in low-altitude monitoring frameworks, target-tracking frameworks, precise guidance frameworks and forest fire prevention frameworks [1–3]. From one perspective, the grayscale information of the targets is concentrated in a small area in images, lacking shape and further information [4–6]. Then again, the difference between the objects and its encompassing background clutter is normally low [7–11]. Hence, these unfavorable factors make infrared small-target detection a difficult course. Therefore, the field of IR small-target detection has received a lot of attention [12–14]. Quantities of the most recent strategies for IR small target have been proposed.

Currently common IR small-target detection methods can mainly be divided into two categories: (1) background suppression-based methods; (2) matrix reconstruction-based methods; (3) human visual system-based methods.

The method based on background suppression refers to methods of transforming an IR image in spatial or transformation domain. The top hat method [15] is fast but sensitive to clutter, and the maximum median and maximum mean filter [16] is also widely used to suppress background clutter. Shao [3] uses a Laplace–Gaussian (LoG) filter to enhance image contrast. Wang [17] proposed a difference of Gaussian (DoG) operator similar to LoG but easier to construct. Han [18] used the Gabor operator instead of the Gaussian operator and added directionality to the operator, then he proposed IDoGb method. Qi [11] uses the facet model to derive the second derivative of the row and column directions and proposes a fast saliency method. Guo [19] proposed a method using phase spectrum of quaternion Fourier transform. Boccignone [20] applies the wavelet transform to the infrared small-target detection field. The spatial-based method is simple and efficient. However, these methods

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cannot be adapted to many situations and do not work well due to complex background.

The matrix reconstruction-based method assumes that the actual image is a linear superposition of the target image, the background image and the noise image. It obtains the target image by optimizing the objective function. The infrared-patch-image (IPI) model proposed by Gao [7] provides the basis for later research; He [21] applied the low rank representation [22] to the field and proposed the LRSR method; Dai used the local steering kernel as the penalty coefficient and proposed the WIPI method [23], he introduces the concept of tensor and proposes the RIPT model [24]; Wang introduces the method of multi-word space learning, and proposes the SMSL method [25], introducing the total variation model as the penalty coefficient, and proposes TV-PCP model [26]. This type of method uses the optimization method to solve the target image. Its running speed is slow, and it is difficult to meet the requirements of real-time performance. At the same time, there are many parameters to set, and the robustness of the algorithm cannot be guaranteed.

In recent years, IR small-target detection methods based on human visual system (HVS) have attracted a large number of scholars. As indicated by the HVS, the difference between the objects and the background enables human to recognize the objects [17–27]. Based on this theory, Chen et al. [8] proposed a local contrast method (LCM). Later, Han [9] proposed an improved local contrast method (ILCM) and intended to increase the speed of the algorithm by setting the step size to half of the window size. In 2016, Chen [28] proposed a method to get a saliency map (LSM), Qin [29] achieved multi-scale by limiting the number of pixels involved in the calculation of the mean, and proposed a novel local method (NLCM) method and Deng [30] proposed a weighted local difference measure (WLDM). Han [10] proposed a multi-scale relative local contrast method (RLCM), Bai [31] introduces the derivative entropy and propose the derivative entropy-based contrast measure (DECM). Liu [32] introduced the concept of flux and proposed a method based on flux density and direction diversity in gradient vector field (WMFD). Shi [11] proposed a high-boost-based multi-scale local contrast measure (HB-MLCM), which greatly improved the detection rate. The methods based on HVS are more scientific and effective. However, these approaches still cannot solve the problems such as ocean–sky background, linear interference, and point-like highlight noises.

In addition to the above three types of methods, there are methods of non-parametric regression [33], multi-feature analysis method [34], directional high-pass filter based on LS-SVM method [35] and deep learning method [36] have been proposed. However, these methods have large calculations or many parameters, resulting in poor robustness, low detection efficiency, and poor accuracy. Therefore, they are not receiving much attention.

This paper proposes an infrared small-target detection strategy based on multi-directional multi-scale high-boost response (MDMSHB). Considering that most of the background edges will extend in a certain direction, an eight-direction filtering template based on HVS is proposed. The template can effectively suppress heterogeneous background such as cloud, linear interference and interface like ocean–sky background. Then a map based on multi-directional multi-scale high-boost response is calculated. Finally, a straightforward threshold segmentation technique is utilized to get the detection result. The simulation results comparing this method with the four state-of-the-art strategies in six sequences demonstrate that the proposed strategy can adequately suppress heterogeneous background and arbitrary noise. The approach can improve detection rate and reduce false alert rate as well when dealing with circular targets and elliptical targets.

This paper is organized as follows: Sect. 2 presents the HB-MLCM algorithm; Sect. 3 introduces the MDMSHB algorithm flow and threshold segmentation method in detail, and analyzes the feasibility of the method; Sect. 4 demonstrates the experiment and results; the conclusion of this paper is given in the Sect. 5.

2 Methodology

2.1 Directional filters

During image-processing steps, the numerical features (such as mean and variance) of the area are calculating all pixels. Considering that most of the background edges will extend in a certain direction, this computing strategy seriously ignores the direction information in the image. And it is insensitive to clutter with obvious dividing lines (such as sea and sky background). Therefore, as in Eqs. (1)–(4), we have designed an eight-direction filter template Filter$(2n+1, 2n+1)$ with a filter size of $[2n+1, 2n+1], n = \{1, 2, \ldots\}$. The filter template is defined as follows:

$$\text{Filter}_\alpha = \{f_{ij}\}, f_{ij} = \begin{cases} \frac{1}{(n+1)(2n+1)} & (i,j) \in A_1 \\ 0 & (i,j) \in A_2, \end{cases} \tag{1}$$

where $\alpha = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$ and α represents the direction of the filter, and (i, j) denotes the position of the current pixel. The two areas A_1 and A_2 are defined as follows:

$$A_1 = \{(x, y) | \max(\Delta x, \Delta y) \le n, \ell(x, y, \alpha) \ge 0\}, \tag{2}$$

$$A_2 = \{(x, y) | \max(\Delta x, \Delta y) \le n, \ell(x, y, \alpha) < 0\}, \tag{3}$$

where x and y represent the number of rows and columns of the pixel at the filter position, respectively. $\Delta x = |x - n - 1|$ and $\Delta y = |y - n - 1|$ represent the distance

of the point (x, y) from the center point on the vertical and horizontal coordinates, respectively. The definition of $\ell(x, y, \alpha)$ is as follows:

$$\ell(x,y,\alpha) = \begin{cases} (-1)^{\frac{\alpha}{\pi}}(n+1-y) & \alpha = 0 \text{ or } \pi \\ (-1)^{\frac{\alpha}{\pi}-\frac{1}{4}}(x-y) & \alpha = \frac{\pi}{4} \text{ or } \frac{5\pi}{4} \\ (-1)^{\frac{\alpha}{\pi}-\frac{1}{2}}(x-n-1) & \alpha = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} \\ (-1)^{\frac{\alpha}{\pi}-\frac{3}{4}}(x+y-2n-2) & \alpha = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}. \end{cases} \tag{4}$$

Figure 1 shows eight filter templates for $n = 2$. The "1" in the template indicates the pixel position participating in the calculation. It is easy to think that due to the consideration of information in all directions, the background will be greatly suppressed when dealing with the ocean–sky background. And the processing result tends to zero. The eight-direction filter template can effectively suppress the background clutter, and improve the signal-to-clutter ratio. And this strategy support parallel computing. It can improve the detection effect while ensuring efficiency.

2.2 Directional high-boost filtering

High-boost filtering is a classic image-preprocessing method. This method can preliminary enhance the detail information of an image. It is widely applied in painting and publishing to sharp an image. In this paper, considering the pixel values in an image are all positive, we set the entities which are less than zero in the result of high-boost filtering to zero [7]. Therefore, the first step of MDMSHB is to carry out the directional improved high-boost filtering according to the following equations:

$$\hat{I}_{hpi} = \max\{I_0 - I_{\mathrm{mi}}, 0\}, \quad i = 1, 2, \ldots, 8, \tag{5}$$

$$I_{\mathrm{IHBFi}} = I_0 \cdot \hat{I}_{hpi}, \tag{6}$$

where i denotes the ith direction of filters; I_0 represents the original image; I_{mi} and I_{IHBFi} represent the smooth image and the improved high-boost image in the ith direction; $\cdot$ denotes the multiplication of the corresponding elements.

2.3 Multi-direction and multi-scale operation

The scale-adaptive process of local contrast means that some significant areas suitable for the target size can be extracted from multiple scales for further processing, which helps to improve detection accuracy. In this paper, the local contrast calculation strategy is adopted. As shown in Fig. 2, the red box indicates the target region, and the area between the white box and the red box are the background region [7].

During the processing steps, we slide the windows from the left-top to the right-bottom of an image while changing size of the target and background region at each pixel location to realize multi-scale operation. This method takes multi-direction and multi-scale information into account. It can obtain great recognition rate, meanwhile effectively suppressing heterogeneous background such as cloud, linear

(a)

1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
1	1	1	0	0

(b)

0	0	0	0	0
0	0	0	0	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

(c)

0	0	1	1	1
0	0	1	1	1
0	0	1	1	1
0	0	1	1	1
0	0	1	1	1

(d)

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
0	0	0	0	0
0	0	0	0	0

(e)

1	0	0	0	0
1	1	0	0	0
1	1	1	0	0
1	1	1	1	0
1	1	1	1	1

(f)

0	0	0	0	1
0	0	0	1	1
0	0	1	1	1
0	1	1	1	1
1	1	1	1	1

(g)

1	1	1	1	1
0	1	1	1	1
0	0	1	1	1
0	0	0	1	1
0	0	0	0	1

(h)

1	1	1	1	1
1	1	1	1	0
1	1	1	0	0
1	1	0	0	0
1	0	0	0	0

Fig. 1 The eight-direction filtering template for $n = 2$, **a**–**h** are $\alpha = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

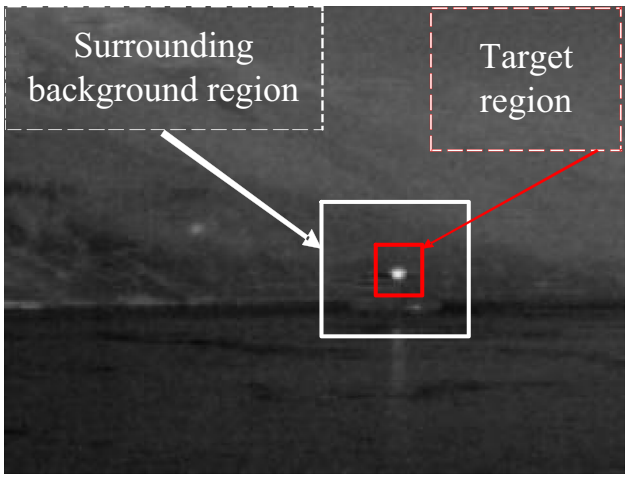

Fig. 2 Definition of target region and background region

interference and interface like ocean–sky background. A series of processing is performed on the directional high-boost filtering result to obtain the result. The specific steps of the strategy are shown as follows:

$$d_i^k = m_{Ti}^k - m_{Bi}^k, \quad k = 1, 2, \ldots, K, \tag{7}$$

$$m_{\mathrm{Ti}} = \frac{1}{N_{\mathrm{Ti}}} \sum_{p=1}^{N_{\mathrm{Ti}}} \upsilon_{\mathrm{p}}, \quad m_{\mathrm{B}} = \frac{1}{N_{\mathrm{Bi}}} \sum_{q=1}^{N_{\mathrm{Bi}}} \upsilon_{\mathrm{q}}, \tag{8}$$

$$C_i = \left(\max_{k \in \{1,2,\ldots K\}} \left(\left| d_i^k \right| \right) - \min_{k \in \{1,2,\ldots K\}} \left(\left| d_i^k \right| \right) \right)^2, \tag{9}$$

where i and k denotes the ith direction and the kth scale; d represents the local difference; m_{Ti} and m_{Bi} represents the pixel mean to participate in calculation of the target and background region in the ith direction, respectively; similarly, N_{Ti} and N_{Bi} represents the number of pixels participating in calculation in the two region. υ_{p} and υ_{q} represent the values of the pth and qth pixels participating in the calculation, respectively.

According to Eqs. (7)–(9), we can obtain the response in the ith direction C_i where $i = 1, 2, \ldots, 8$. In other words, eight local contrast maps $C_i, i = 1, 2, \ldots, 8$ can be calculated in eight directions. In the MDMSHB map (denoted by C), a higher value demonstrates a higher likelihood that there is a small target. To combine these local contrast map and incorporate helpful information, the fusion method of taking the minimum value is adopted as follows:

$$C = \min_{i \in \{1,2,\ldots,8\}} C_i. \tag{10}$$

As indicated by the steps over, the detection process is given by Algorithm 1. M and N represent the size of the IR image, and the selection of the K value will be explained in Sect. 3. Figure 3 shows an example of MDMSHB processing.

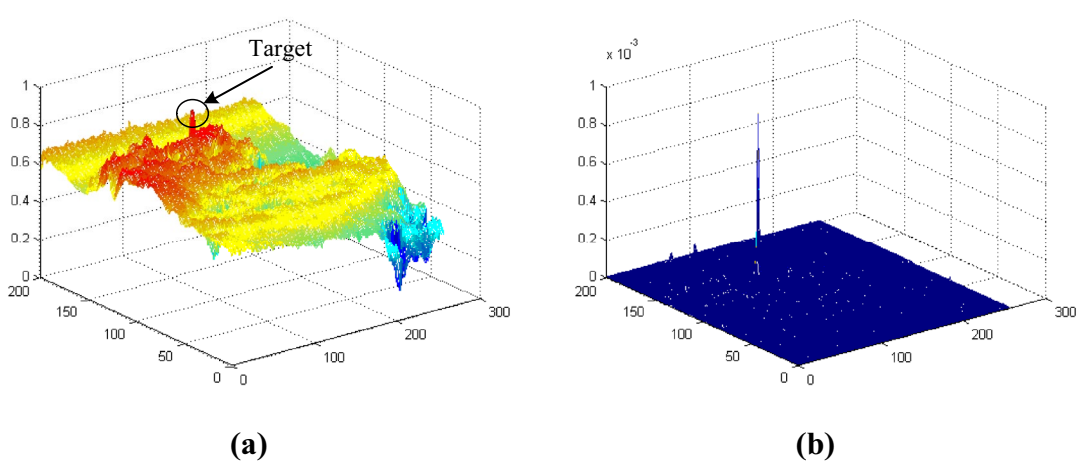

Fig. 3 MDMSHB processing example. **a** Original image 3D display. **b** MDMSHB map

Algorithm 1 Infrared Small Target Detection Based on Multi-directional Multi-scale High Boost Response

Input: IR image I
Output: MDMSHB map

```
for ii=1:M do
    for jj=1:N do
        for i=1:8 do
            Calculate the high boost image I_IHBFi in the i th direction
            for k=1:K do
                d_i^k = m_Ti^k − m_Bi^k
            end for
            C_i = ( max_{k∈{1,2,···K}} (|d_i^k|) − min_{k∈{1,2,···K}} (|d_i^k|) )^2
        end for
        C = min_{i∈{1,2,···,8}} C_i
        Replace the current center pixel value with the C
    end for
end for
return MDMSHB map
```

2.4 Threshold segmentation

In this paragraph, we introduce an adaptive threshold segmentation method. We obtained the MDMSHB map in the previous section. Next we use the method to obtain the target position. The threshold T can be expressed as:

$$T = \mu + \eta * \sigma, \tag{11}$$

where μ and σ represent the mean and variance of the MDMSHB map, respectively. The symbol * indicates multiplication. And η is the adjustment factor, which is generally set to 4–7. Figure 4 shows the operation flow of the algorithm in four directions.

3 Experiment

3.1 Experiment setting

(a) Evaluating indicators

Signal-to-clutter ratio gain (SCR Gain) and background suppression factor (BSF) are broadly utilized for quantitative assessment in small-target detection. Their expression is defined as:

$$\mathrm{SCRG} = \frac{(S/C)_{\mathrm{out}}}{(S/C)_{\mathrm{in}}}, \quad \mathrm{BSF} = \frac{C_{\mathrm{in}}}{C_{\mathrm{out}}}, \tag{12}$$

where S denotes the signal amplitude, C denotes the standard deviation of the clutter, and in and out, respectively, represent the input IR image and the output MDMSHB map. As can be seen from the above definition, the better the impact of small target prominence, the higher the value of SCRG, in the same way, the better the background stifling impact, the higher is the value of BSF.

Likewise, receiver-operating characteristic (ROC) curve is additionally regularly utilized for quantitative evaluation. It depicts the dynamic association between false-positive rate (FPR) and true-positive rate (TPR). Generally speaking, the closer the ROC curve is to the upper left corner, the larger the area under the curve (AUC), which also shows the higher the detection accuracy of the algorithm. Compared with numerical measures, the ROC curve can more readily show the performance of these algorithms. The definitions of TPR and FPR are given by

$$\mathrm{TPR} = \frac{\%\ \text{real targets detected}}{\%\ \text{real targets}}, \tag{13}$$

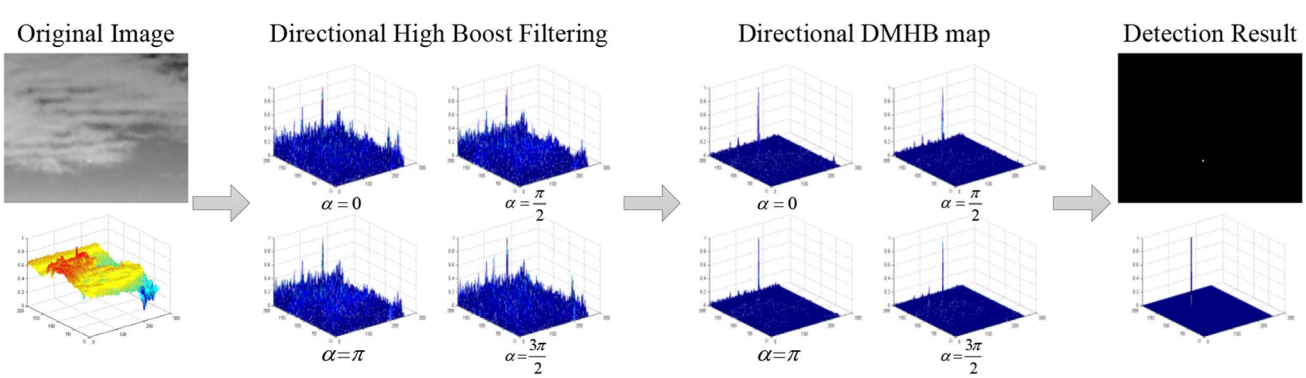

Fig. 4 MDMSHB method flowchart

$$\mathrm{FPR} = \frac{\%\ \text{number of pixels detected in background region}}{\%\ \text{real targets}}. \tag{14}$$

(b) Test data

In the experimental part, we tested six sequences, totaling 1142 images. Among them, cloud background, noise interference, weak contrast and other characteristics will make it difficult to detect infrared small targets. The resolution, target features and background features of the sequence are described in detail in Table 1. The sequence sample will be shown in the next section. In addition, to illustrate the robustness and detection effect of MDMSHB more intuitively, we have tested four extra complex images in the background, which will be shown in the next section.

All the calculations in this paper are performed on a computer with i5 7500 CPU, 3.4 GHz, and 8 Gb of memory. The software is MATLAB R2014a.

3.2 Experimental results

In the experiment, the multi-scale number K is set to 4, the multi-scale target region diameters are 3×3, 5×5, 7×7 and 9×9; the background region diameter is set to 15×15, the adjustment factor η is set to 7.

In the experiment, so as to successfully check the execution of the proposed strategy, we show six real infrared image sequences with typical background. Figure 5 shows the results of different steps of the MDMSHB method. As shown in Fig. 5a–f are the original infrared image, the original image 3D display, the first direction high-boost filtered result, the MDMSHB map, the MDMSHB map 3D display and the detection result.

The detection process of row 1–6 in Fig. 5 is sequence 1–6, respectively. To better display the position of the object in the image, we frame the object in a red box in two-dimensional graph and circle the object in a red circle in three-dimensional graph. Sequence 1 is an aerial and astronomical background, and the target is disturbed by transverse clouds, and the image is strongly affected by noise; sequence 2 is a long strip with structured clouds at the top; clouds in sequence 3 are circularly distributed around the target; SEQUENCE 4 has two targets, and the background is relatively pure, but the contrast between the target and the background is very low; sequence 5 has an arc-shaped structure in the upper left corner. The background of sequence 6 is very complex, the target is buried behind a large number of clouds, and the moving range of the target in the sequence is large.

Figure 5 demonstrates that the proposed strategy can accomplish great recognition execution for small targets. The homogeneity background clutter is well stifled by the proposed strategy. In addition, when the proposed method handles the heterogeneous background, it can still effectively suppress the background and extract the target precisely.

To additionally confirm the viability of the proposed strategy, we utilize four different strategies for examination. These methods are top hat transform (Tophat) [15], maximum median method (MaxMedian) [16], local contrast method (LCM) [8] and high-boost-based multi-scale local contrast measure (HB-MLCM) [7]. Figure 6 shows the results of sequence 1–6 processed by the above five methods. Figure 7 is a three-dimensional display of a two-dimensional image at the corresponding position in Fig. 6.

From Figs. 6 and 7, we can intuitively compare the advantages and disadvantages of various methods. Similar to Fig. 5, targets in two-dimensional images are framed in red boxes and targets in three-dimensional images are circled in red circles. There are much clutter in Tophat method and MaxMedian method. MaxMedian method is even difficult to locate the target in Sequence 1. The LCM method only roughly enhances the original image

Table 1 Description of test sequence

Sequence	Resolution	Target description	Background description
Seq 1	128×128	Targets are dotted, with a total of 9 pixels	The image resolution is low, with lateral cloud disturbance and a lot of random noise
Seq 2	256×200	Targets are long strips, which occupy more pixels	The background is relatively pure and there is lateral cloud interference above the image
Seq 3	320×240	The target is roughly circular, on the left side of the image	On the right side of the image, there are a large number of clouds, which are circularly distributed around the target
Seq 4	208×152	There are two in the image, one is darker and the other is brighter, both located in the middle of the image	The background is relatively pure, but there is little difference between the target and the background
Seq 5	632×504	The target is circular, located on the left side of the image	The image has high noise and arc-shaped structural clutter in the upper left corner
Seq 6	256×172	The target is a long strip, with the middle part highlighted	The background is very complex and the target is buried in the clouds

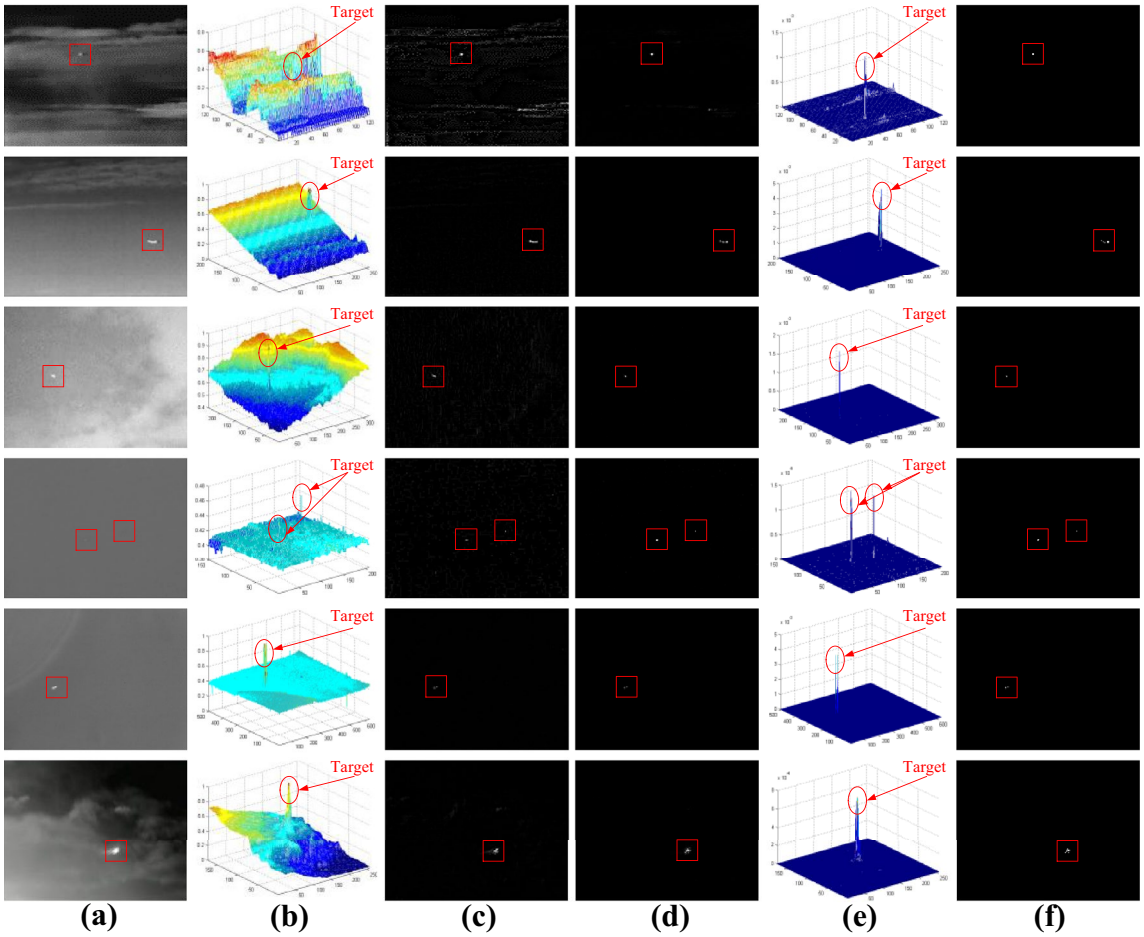

Fig. 5 Single image-processing results. **a** Original infrared image; **b** original image 3D display; **c** high-boost filtered results in the first direction; **d** MDMSHB map; **e** MDMSHB map 3D shows; **f** threshold segmentation result

and does not obviously suppress the background. The HB-MLCM method can detect the target better, but the background suppression is not very good. Obvious clutter can be observed in sequences 1 and 4. From the experimental results in the figure, we can see that MDMSHB method can not only detect the target accurately, but also suppress the background better, making the result more pure.

To better illustrate the scenario adaptability of the MDMSHB method, we tested four additional complex scenarios. The original image and the test results of various methods are shown in Fig. 8, and Fig. 9 shows the three-dimensional image corresponding to Fig. 8. The marking rule is the same as the figure above.

Scene 1 is an aerial-sky background, the target is buried in a large number of clouds, and the upper part of the image has many structural clouds; scene 2 is a sea–sky background, the target occupies less pixels, the image has sea level in the middle, and there is a highlight bridge body interference; scene 3 is also the sea–sky background, the target is on the left side, there is a bright lighthouse disturbance on the right side, and there is a boat similar to the target feature in the middle; the lower part of scene 4 is the sea and the upper part is the mountain. There are irregular textures in the mountains and there is highlight noise.

From Figs. 8 and 9, we can see that the Tophat method and MaxMedian method are not effective because of their simple principles. Although LCM applies the concept of human visual system, its calculation is relatively simple. It only uses maximum and mean values, and its detection effect is not ideal. Both HB-MLCM and MDMSHB methods have been processed by high-boost filtering, which enhances the target characteristics and uses more scientific saliency map calculation method. Therefore, the effect of the two methods is superior to that of other methods. However, MDMSHB method takes into account that clutter will extend in a certain direction, and adds the concept of directional filtering. Therefore, its detection results can better suppress structural clutter, such as bridges, clouds,

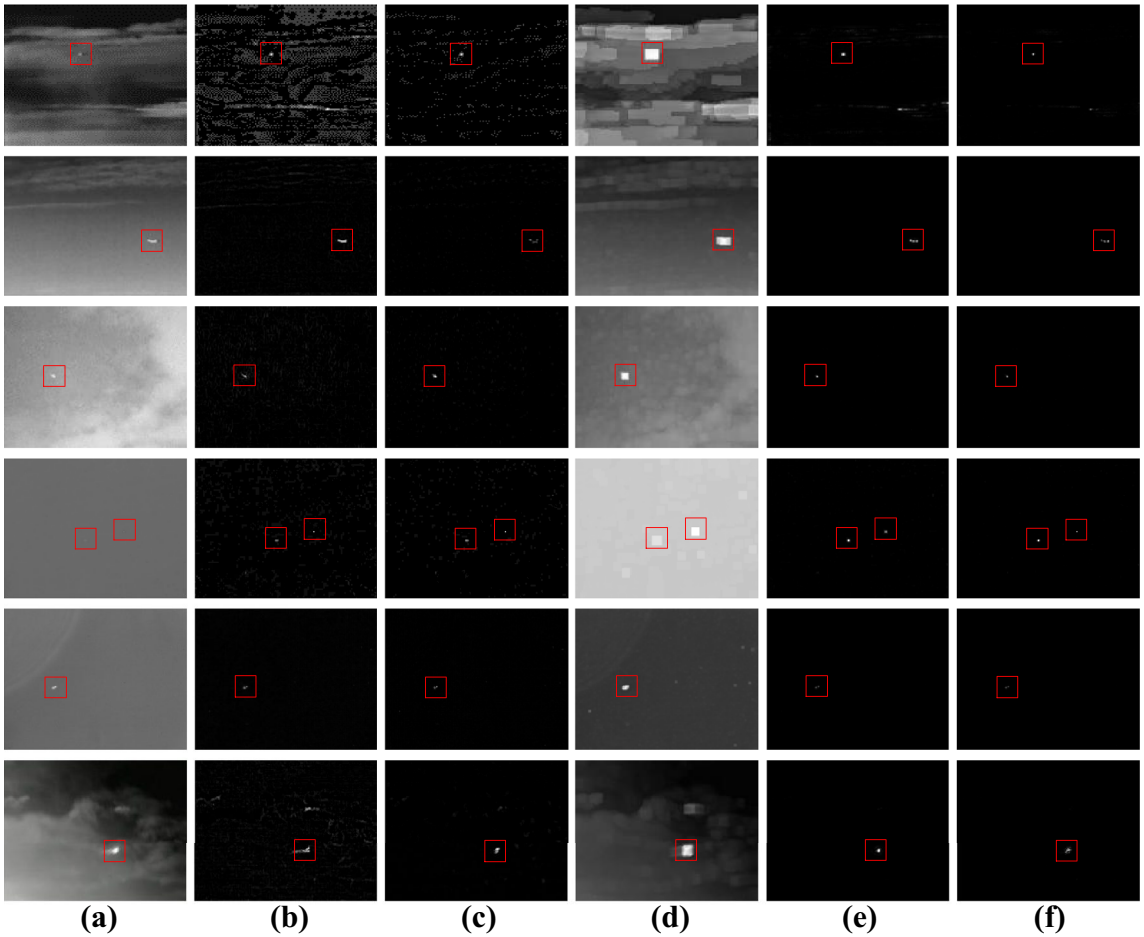

Fig. 6 The detection result of five algorithms. **a** Original image; **b** result of Tophat; **c** result of MaxMedian; **d** result of LCM; **e** result of HB-MLCM; **f** result of MDMSHB

lighthouses and so on. It can also be seen from the display results in the figure that MDMSHB can get the best detection results, and its background clutter is the least.

The above experiments demonstrate the effectiveness of MDMSHB method from a qualitative point of view. The following will illustrate the excellent performance of MDMSHB method from a quantitative evaluation point of view.

This paper quantitatively compares the results of the images in sequences 1–6 with the five algorithms tested. The results of SCR Gain and BSF are shown in Table 2, where the maximum value is marked in red.

It can be seen from the results in Table 2 that the MDMSHB method reaches a maximum in all sequences, which means that the MDMSHB method can not only detect the target, but also greatly suppress the background. More pure results can be obtained compared to other methods. For this evaluation parameter, the MDMSHB method performs better than others.

In addition, as shown in Fig. 10, we tested the ROC curves of six sequences. This can further demonstrate the superior performance of the MDMSHB method. As can be seen from the figure, the curve of the MDMSHB method can always be closer to the upper left corner, especially in the sequences 3–5, and at the same time, the excellent levels can be achieved in the sequences 1, 2, and 6.

To accurately demonstrate the advantages of the MDMSHB method in terms of ROC curve, we recorded the AUC of the above method in Table 3. Again, the maximum value is marked in red. As can be seen from the table, the MDMSHB method can reach a maximum in the AUC of each sequence, and is ahead of all other methods in the sequences 2–6. The data of the ROC curve can quantitatively explain that the MDMSHB method can lead other methods in detection accuracy.

In the above experimental part, we first show the detection process of the MDMSHB method in the form of pictures, which shows that the method can accurately detect the target. Then, for the sequences 1–6, the comparison of the two-dimensional and three-dimensional images of the five algorithm processing results is shown. It can be

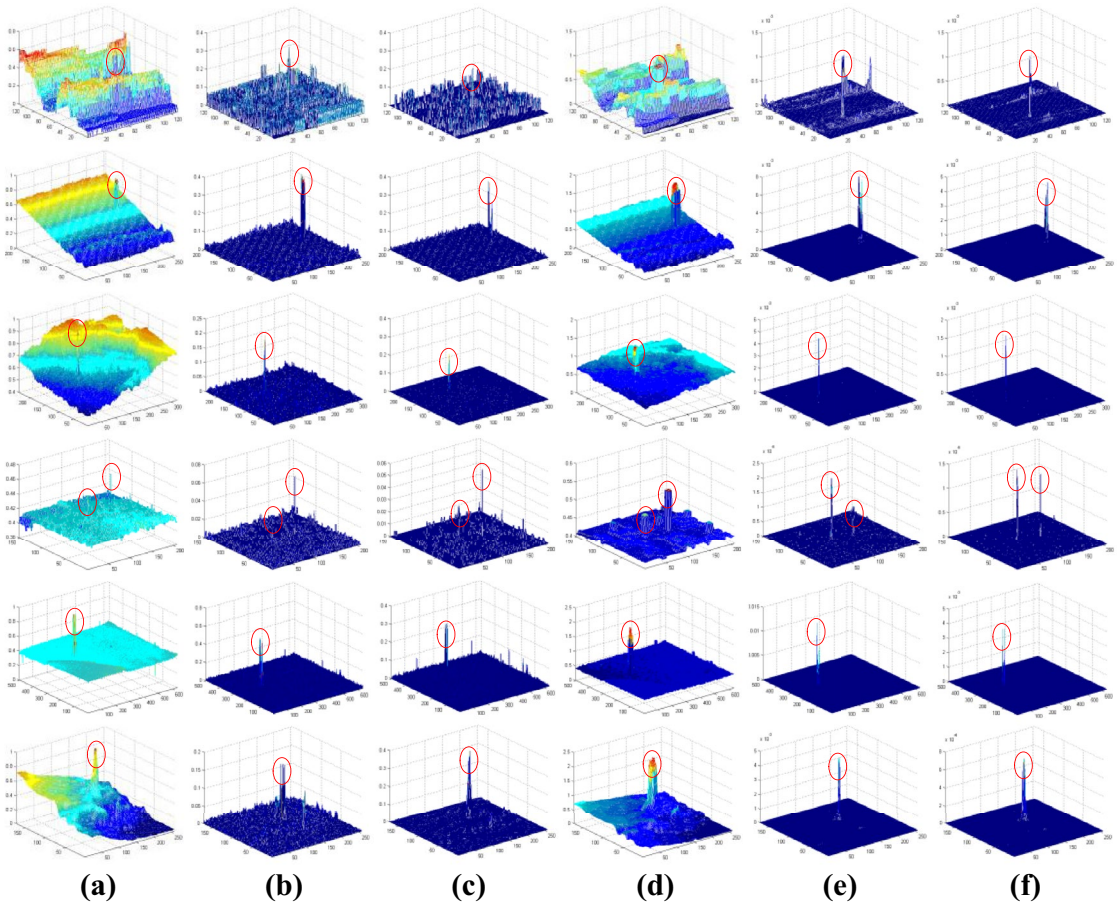

Fig. 7 3D display of Fig. 6. **a** Original image; **b** result of Tophat; **c** result of MaxMedian; **d** result of LCM; **e** result of HB-MLCM; **f** result of MDMSHB

qualitatively seen that the MDMSHB method can accurately detect the target while suppressing the background greatly. Then we tested the images of four complex scenes to further illustrate the scene adaptability and robustness of the MDMSHB method. Next we performed a quantitative evaluation of the sequence for the SCR Gain and ROC curves. As can be seen from the SCR Gain and AUC data in the table, the MDMSHB method can always achieve higher values than other methods. This aspect confirms the qualitative evaluation of the first part, and on the other hand, it also shows that the MDMSHB method can greatly suppress the background and obtain more pure detection results while accurately detecting the target.

4 Discussion

In this section, we will discuss the effectiveness of MDMSHB in different situations. As shown in Fig. 11, when the sliding window slides in the image, it may appear in four situations with the target and noise: entire target region (TR), complete background region (BR), background edge region (BE), and partial target region (PR). Figure 11a shows the difference between the original square filter and the directional filter in this article, Fig. 11b is a concrete representation of the four regions, BR indicates that the filter is all in the background area, and BE indicates that the filter may cross the boundary. PR indicates that the filter value contains a part of the target, but the target center and the filter center do not coincide. TR indicates that the filter just frames the target, and the target center and the filter center are coincident. The contrast mechanism of the HVS means that the difference between the objects and the background enables human to recognize the objects [17–27]. This prior knowledge is useful for finding targets in the entire infrared image and has a high detection rate.

Given an input infrared image I, we calculate the corresponding MDMSHB for each position on each sliding window. After sliding the window through the entire image, the final MDMSHB saliency map is obtained,

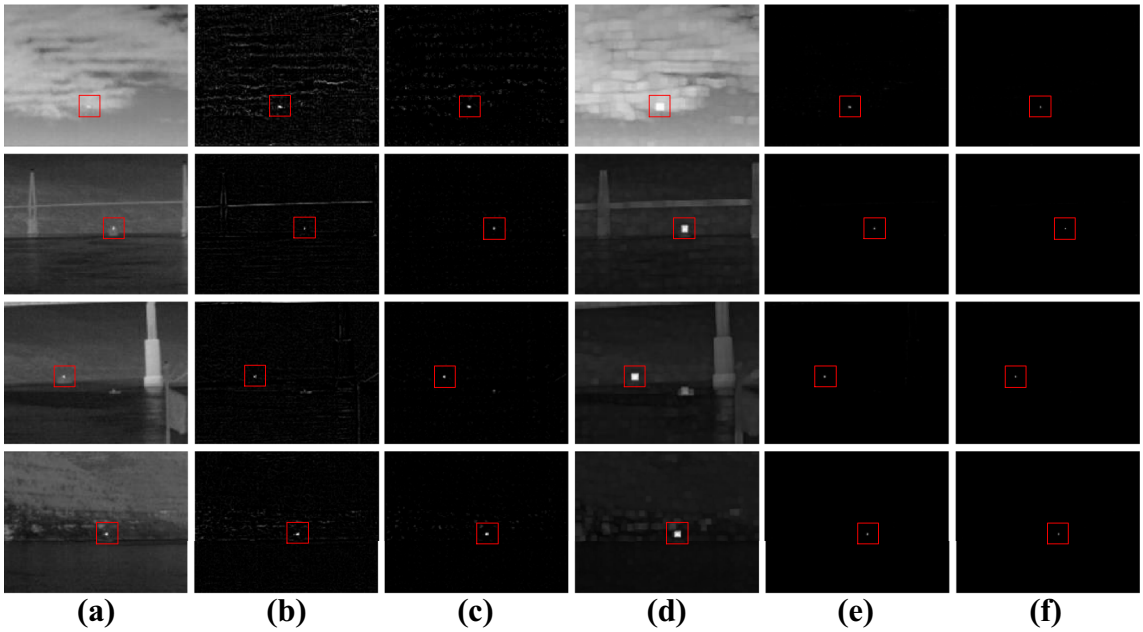

Fig. 8 The detection result of four additional scenarios. **a** Original image; **b** result of Tophat; **c** result of MaxMedian; **d** result of LCM; **e** result of HB-MLCM; **f** result of MDMSHB

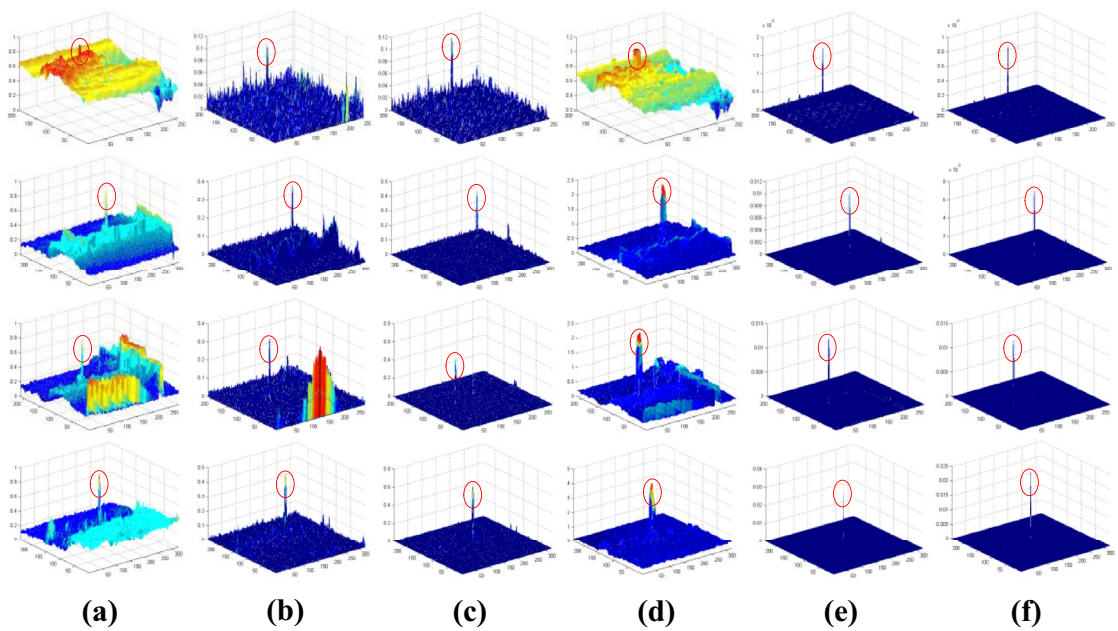

Fig. 9 3D display of Fig. 8. **a** Original image; **b** result of Tophat; **c** result of MaxMedian; **d** result of LCM; **e** result of HB-MLCM; **f** result of MDMSHB

denoted as C. Next, we discuss the corresponding situation of the MDMSHB result when the entire sliding window is located in the above four regions, and the central pixel coordinates are (x, y). In the following formula, let

$$D_i(x,y) = \max_{k\in\{1,2,\ldots K\}} \left(\left|d_i^k(x,y)\right|\right),$$
$$d_i(x,y) = \min_{k\in\{1,2,\ldots K\}} \left(\left|d_i^k(x,y)\right|\right).$$

Table 2 Comparison of SCR Gain and BSF of test sequence

Sequence	Evaluation indicators	Tophat	Max-median	LCM	HB-MLCM	MDMSHB
Seq1	SCR Gain	2.3790	2.0089	1.2505	43.9402	97.3840
	BSF	1.2819	2.9233	0.8321	28.0608	72.6093
Seq2	SCR Gain	8.1702	0.8497	3.0474	329.915	231.010
	BSF	3.3604	3.2367	1.5745	346.647	688.779
Seq3	SCR Gain	3.8877	1.3294	4.7369	191.636	2240.89
	BSF	3.7423	1.9252	3.3504	208.260	3115.42
Seq4	SCR Gain	1.3628	1.8794	2.1003	63.7249	108.966
	BSF	1.1069	1.0864	1.2079	10.0508	27.4875
Seq5	SCR Gain	2.8259	3.8982	1.9524	19.5993	24.5691
	BSF	2.7930	3.6047	1.2888	26.0105	99.7655
Seq6	SCR Gain	11.6805	4.8816	6.0069	6958.95	38,952.8
	BSF	14.1696	12.1815	6.4284	12,424.1	89,415.3

1. When the slide window is in the complete background region (BR), the pixel value of this region will become zero after high-boost filtering, i.e., $D_i(x, y) = d_i(x, y) = 0$. Therefore, for BR, MDMSHB can suppress it to zero without false alarm. The formula is as follows:

$$C_{\mathrm{BR}}(x, y) = \min_{i\in\{1,2,\ldots,8\}} \left\{ \left(D_i(x, y) - d_i(x, y)\right)^2 \right\} = 0. \tag{15}$$

2. When the slide window is located at the background edge region (BE), the special case is when (x, y) is located at the edge, and one direction can always be found in the eight directions, so we can get $D_i(x, y) \approx d_i(x, y) \approx 0$. Therefore, the processing result is the same as BR (as shown in Fig. 11b) or approximately 0. Therefore, MDMSHB can perform better in the face of structural clutter such as edges and lines. The formula is as follows:

$$C_{\mathrm{BE}}(x, y) = \min_{i\in\{1,2,\ldots,8\}} \left\{ \left(D_i(x, y) - d_i(x, y)\right)^2 \right\} \approx 0. \tag{16}$$

3. When the slide window is located at the entire target region (TR), then (x, y) is exactly at the center of the circular target, so the result will be $D_i(x, y) > d_i(x, y)$ regardless of the direction. Obviously, the processing result at this time is greater than zero. The formula is as follows:

$$C_{\mathrm{TR}}(x, y) = \min_{i\in\{1,2,\ldots,8\}} \left\{ \left(D_i(x, y) - d_i(x, y)\right)^2 \right\} > 0. \tag{17}$$

4. When the slide window is located at the partial target region (PR), then (x, y) is not at the center point of the circular target. There will be no small difference between $D_i(x, y)$ and $d_i(x, y)$, but it will be smaller than TR. Thus the MDMSHB result is still greater than zero. Since PR is also the target region, this correspondence is correct. The formula is as follows:

$$0 < C_{\mathrm{PR}}(x, y) < C_{\mathrm{TR}}(x, y). \tag{18}$$

From the above analysis, we can qualitatively know that MDMSHB can better suppress the structural clutter such as edges and lines while accurately detecting the target. Below we will illustrate the advantages of the MDMSHB method over HB-MLCM from an experimental perspective.

As shown in Fig. 12, we compared MDMSHB and HB-MLCM for three additional complex scenarios. In the figure, the original image, the HB-MLCM processing result, and the MDMSHB processing result are from top to bottom, and the black box is left to right in the order of scenes 1–3. The marking method is the same as above.

Scene 1 is an aerial background, and there are a lot of clouds below. The cloud on the right side is very bright, and there is a lot of random noise in the image. The results of HB-MLCM can see that the cloud on the right has a large response and does not suppress the background well. The MDMSHB results in a significant reduction in the right clutter. It greatly suppresses cloud interference while detecting the target. The noise in scene 2 is very strong, there is a circular building disturbance below, and there is a horizontal cloud layer above, the target occupies a small number of pixels and is buried in the cloud. From the experimental results, it can be seen that the HB-MLCM results contain a large amount of clutter, and the buildings in the lower left corner respond greatly, and even the target cannot be accurately detected. However, MDMSHB still greatly suppressed the background, and the building in the lower left corner responded very little. Scene 3 contains a large number of structural clouds and edge information. The target pixel is very small and has little difference with the background. The experimental result shows that HB-MLCM obviously cannot meet the requirements, and there is a lot of clutter at

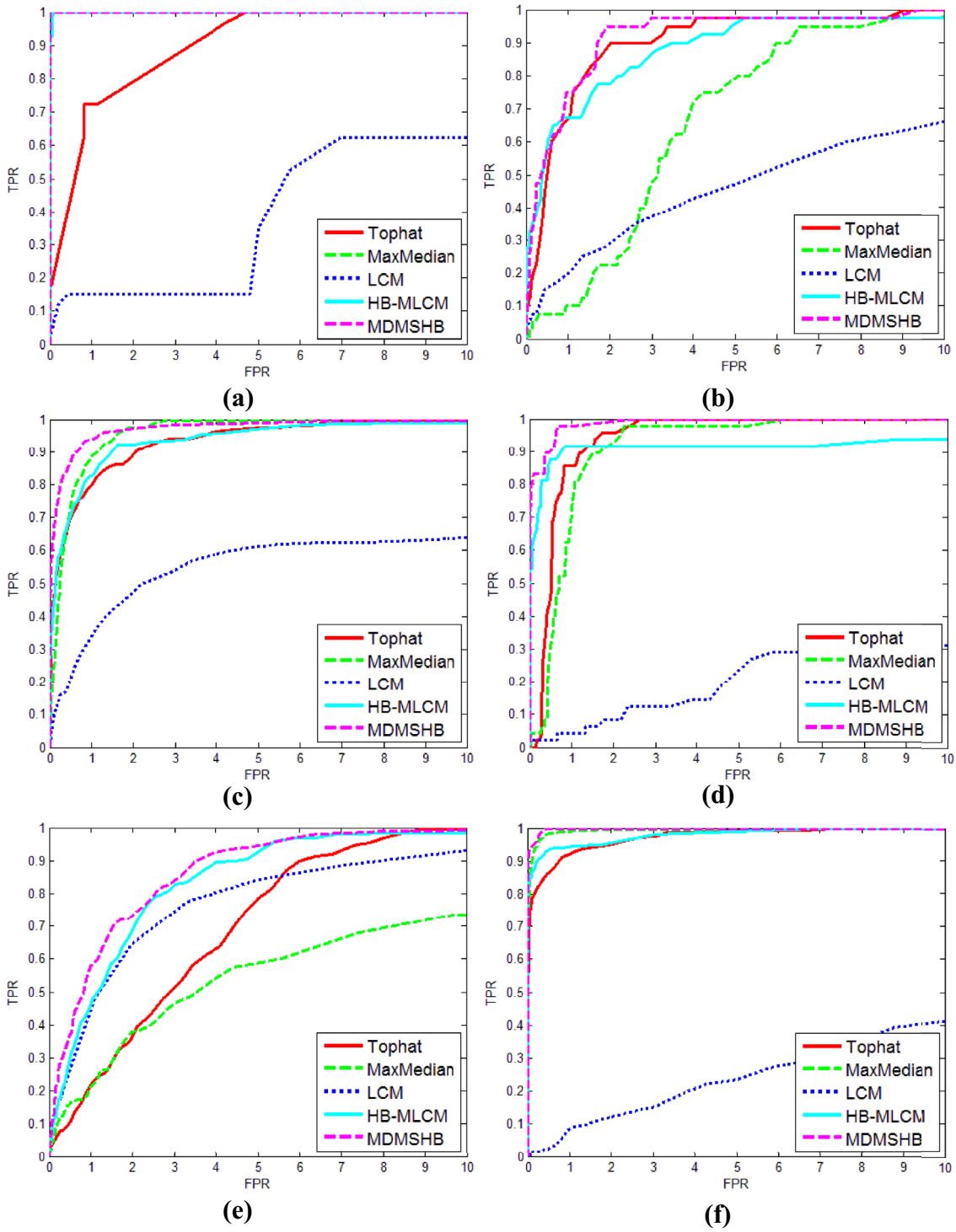

Fig. 10 Comparison of ROC curves for six sequences. **a**–**f** Sequences 1–6

the junction of the upper cloud and the sky. The MDMSHB takes into account the direction information of the structural clutter and can suppress such noise very well, so its clutter is very small.

Next, we compare the MDMSHB method with some machine learning methods. In [35], Wang uses the least squares support vector machine to derive high-pass filtering templates in both directions of rows and columns, and

Table 3 AUC values of the test sequence

	Tophat	MaxMedian	LCM	HB-MLCM	MDMSHB
Seq 1	8.9287	10	3.6734	9.9959	10
Seq 2	8.9662	6.5464	4.4739	8.7662	9.1512
Seq 3	9.2908	9.5289	5.4407	9.3256	9.7118
Seq 4	9.3698	9.0801	2.0113	9.1092	9.8872
Seq 5	6.8263	5.2748	7.5671	8.2729	8.5592
Seq 6	9.7083	9.9293	2.3355	9.7892	9.9812

fuses the two processing results. In the theoretical derivation process, the wavelet kernel is used as the kernel function, and the detection result is obtained by solving the second derivative in the two directions of row and column. It is a background suppression method derived from machine learning methods. Since only high-pass filtering information in two directions is utilized, its algorithm robustness cannot be guaranteed. The processing flow in [35] is to first sample the background image, then generate the target image superimposed with the SNR controlled, and finally use the data to train the deep network for different SNR. This method is based on deep learning, and its accuracy depends on the generated background image and target image. The target style is difficult to estimate in practical applications, and the background is complex and changeable. It is conceivable that the generalization ability of this method cannot be guaranteed. The point of departure of the MDMSHB method and the above two methods is different. Our basis is the prior knowledge of the human visual system. The parameters that need to be determined are only the filter size. Based on this consideration, there are also steps for joint detection of multiple scales. The calculation method in its filter utilizes more information, can perform calculations efficiently, and the introduction of eight directions makes the robustness and generalization ability of the method ensured by considering the edge and other information.

From the above discussion, we can conclude that our method can better suppress heterogeneity background by adding the directional filters and reduce the false alert rate compared with the HB-MLCM algorithm. Besides, this method can obtain higher recognition rate and lower false alert rate while effectively stifling complex edge clutter and arbitrary noise in IR background. At the same time, it has the ability of parallel computing to ensure the detection efficiency.

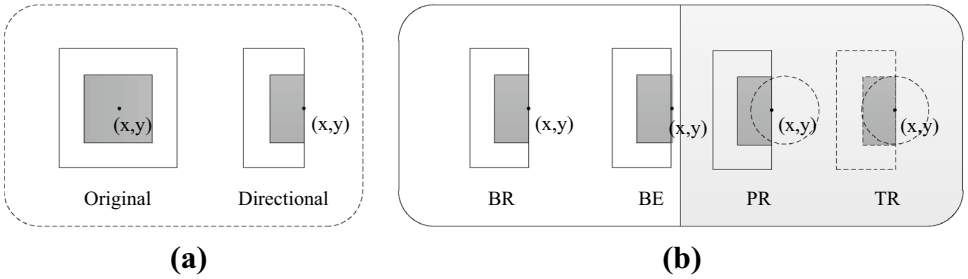

Fig. 11 Filter template and four regions. **a** Schematic diagram of the original filter template and the improved filter template. **b** Schematic diagram of the BR, BE, PR, and TR regions from left to right

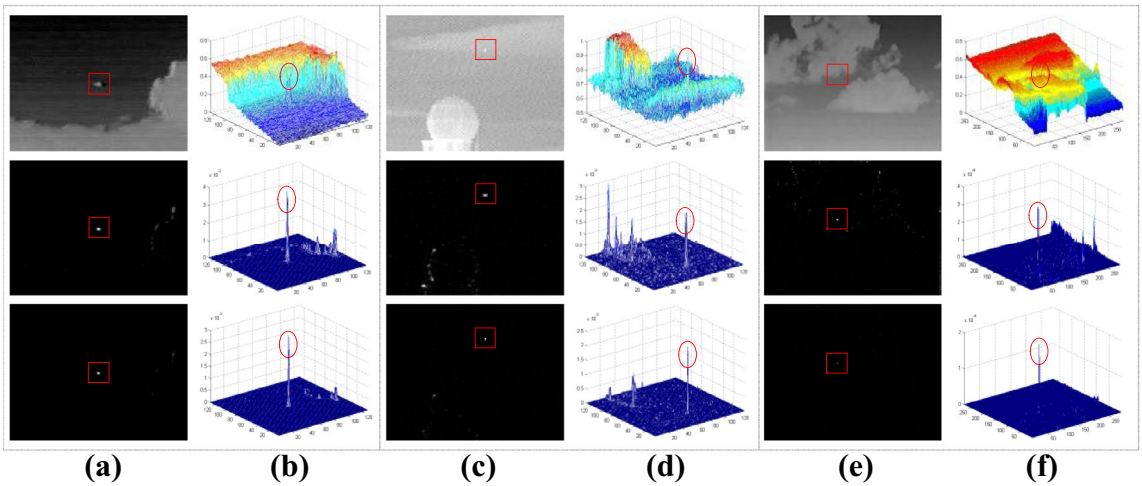

Fig. 12 Comparison of HB-MLCM and MDMSHB. The row from top to bottom is original image, HB-MLCM result and MDMSHB result. **a** Scene 1; **b** 3D display of **a**; **c** scene 2; **d** 3D display of **c**; **e** scene 3; **f** 3D display of **e**

5 Conclusion

In this paper, an infrared small-target detection strategy based on multi-directional multi-scale high-boost response is proposed. According to the characteristics of clutter interference extending in a certain direction, this paper proposes an eight-direction filter to suppress it. Then, the fusion of multiple local contrast maps also proves the robustness and improves the performance of the method. Finally, a straightforward threshold segmentation technique is utilized to get the detection result. This method can effectively suppress heterogeneous background such as cloud, linear interference and interface like ocean–sky background so the target can be effortlessly recognized. As indicated by the experimental results, our method can significantly improve the SCR Gain and BSF. In the actual detection of six real infrared image sequences, our strategy achieved high TPR and low FPR. And the recognition rate of this strategy is superior to the state-of-the-art strategies when dealing with circular targets and elliptical targets. Finally, in the discussion, we also gave a comparison with HB-MLCM and verified the validity of MDMSHB. In summary, the method proposed in this paper is an effective one in the IR small-target detection field.

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Compliance with ethical standards

Conflict of interest On behalf of all the authors, the corresponding author states that there is no conflict of interest.

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